

Nonlinear ILC

Avoiding noise amplification through nonlinear filters

Leontine Aarnoudse

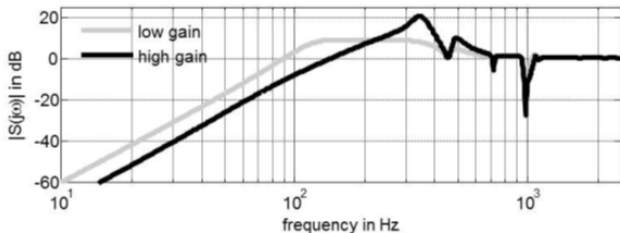
Department of Geoscience
Norwegian University of Science and Technology

1. Nonlinear ILC: main idea
2. Non-repeating disturbances in ILC
3. Nonlinear learning and design
4. Simulation example
5. If time permits: convergence/stability for nonlinear ILC/RC

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- Use a nonlinear ILC controller to increase design freedom

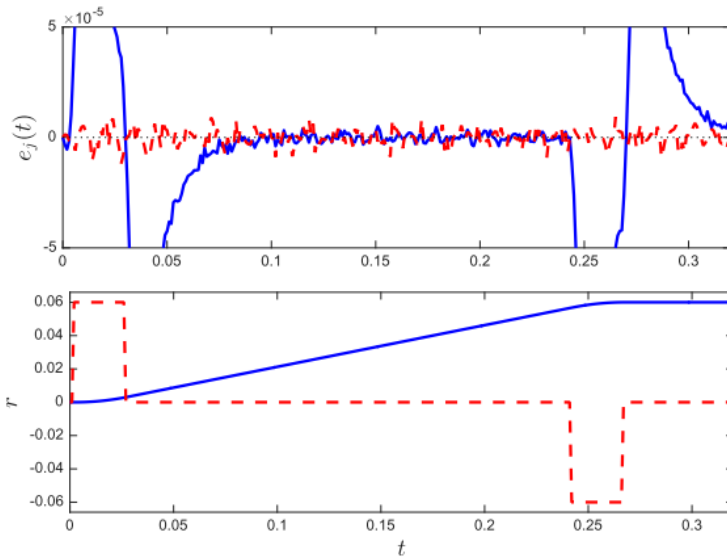
- ▶ Use a nonlinear ILC controller to increase design freedom
- ▶ Inspired by idea of variable-gain feedback control



Which sensitivity for which input?

- ▶ input at 10 Hz
- ▶ input at 350 Hz

And what if the disturbance spectrum is time-varying, e.g., acceleration and constant velocity?



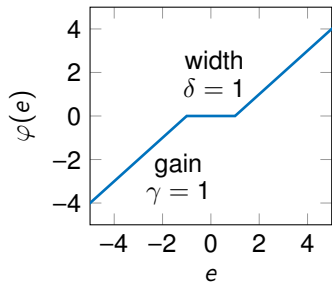
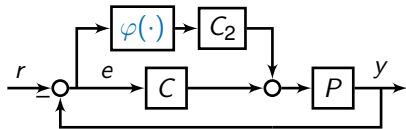
Error and reference signals
wafer stage application

[Oomen and Rojas (Mechatronics 2017)]

Top: error without
feedforward
error with feedforward

Bottom: reference
acceleration

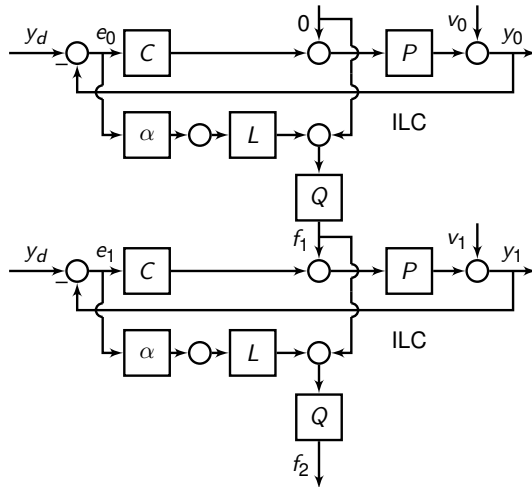
Feedback: low gain
between 0.1 and 0.24, high
gain elsewhere

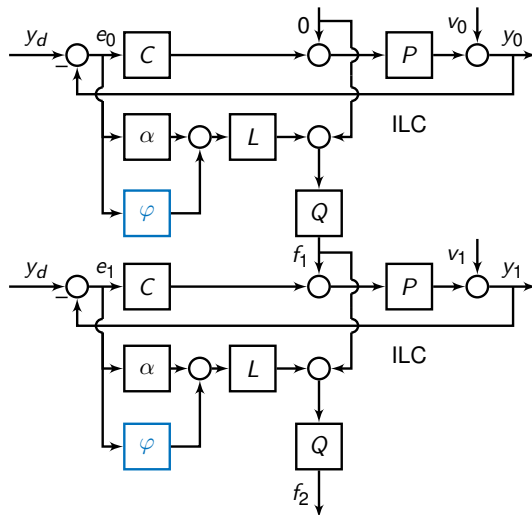


Extra controller C_2 and deadzone nonlinearity $\varphi(e)$:

$$\varphi(e) = \begin{cases} 0, & \text{if } |e| \leq \delta \\ \left(\gamma - \frac{\gamma\delta}{|e|}\right) e, & \text{if } |e| > \delta. \end{cases}$$

Key idea: apply C_2 only for error signals that exceed δ





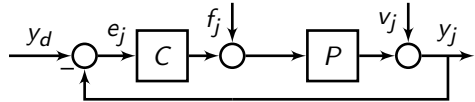
But why do this for ILC?

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$$e_j = \underbrace{\frac{1}{1+PC}}_S (y_d - v_j) - \underbrace{SP}_J f_j$$

ILC can compensate:

- ▶ everything that repeats (Sy_d)
- ▶ nothing that doesn't repeat (Sv_j)



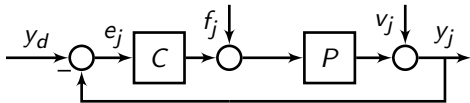
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Through n_e experiments with $f_j = 0$:

$$\widehat{Sy_d} = \frac{1}{n_e} \sum_{j=0}^{n_e-1} e_j$$



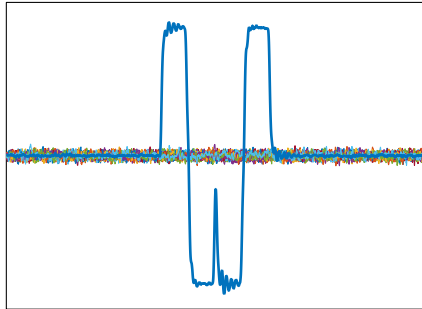
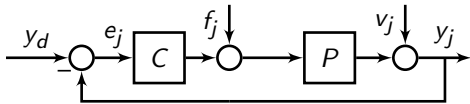
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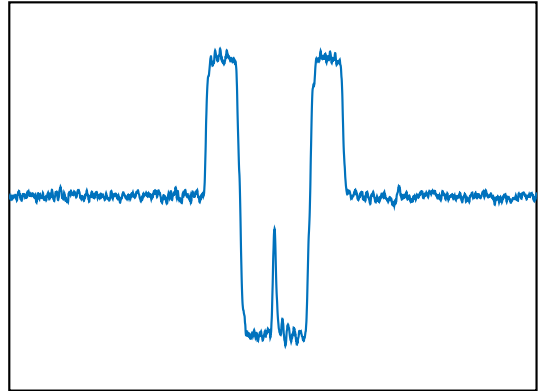
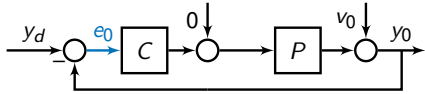
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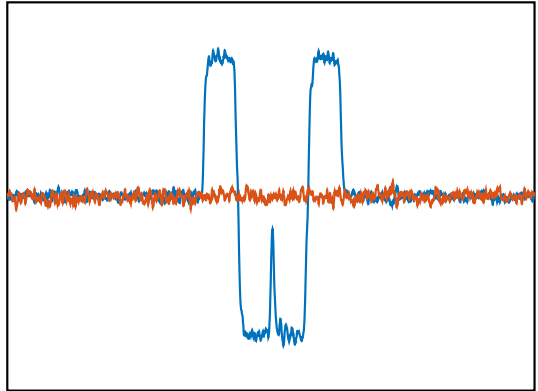
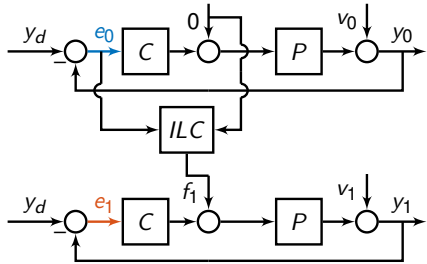
$$\widehat{Sy_d} = \frac{1}{n_e} \sum_{j=0}^{n_e-1} e_j \text{ and } \widehat{Sv_j} = \widehat{Sy_d} - e_j$$





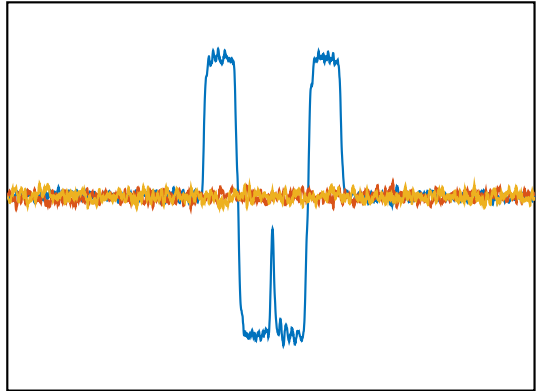
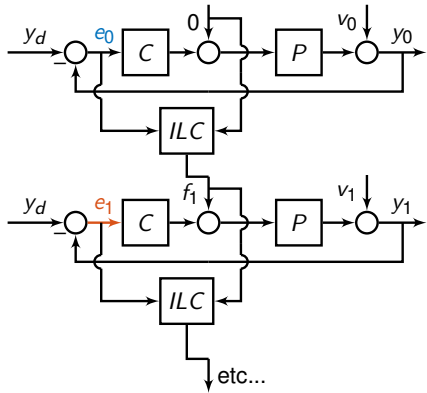
Iterative learning control compensates repeating errors...

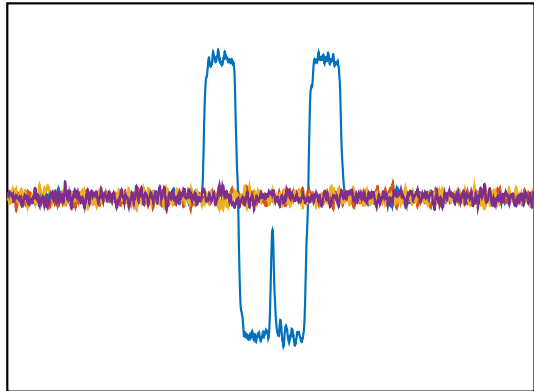
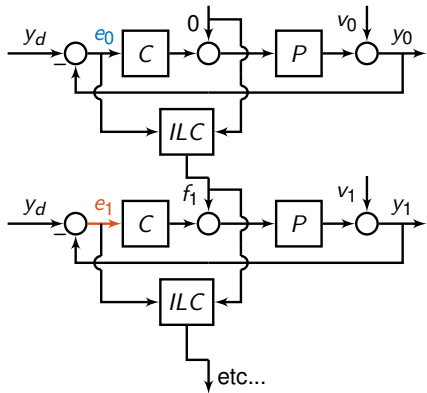
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Iterative learning control compensates repeating errors...

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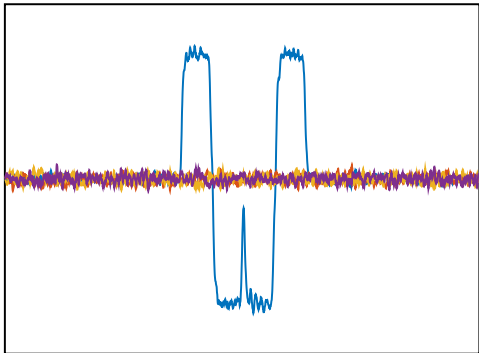




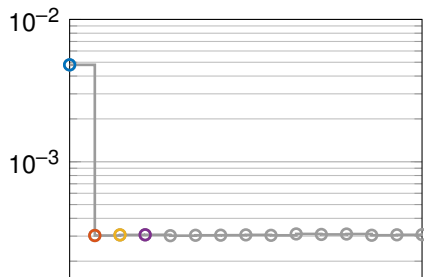
No repeating error after one iteration!

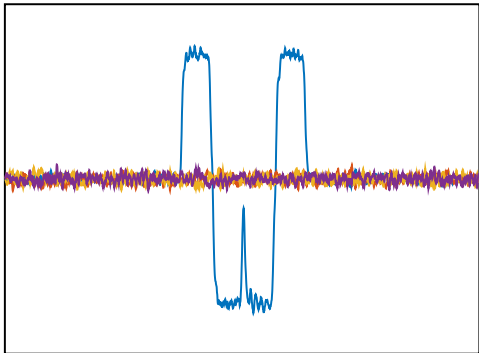
... yet iterative learning control also amplifies noise!

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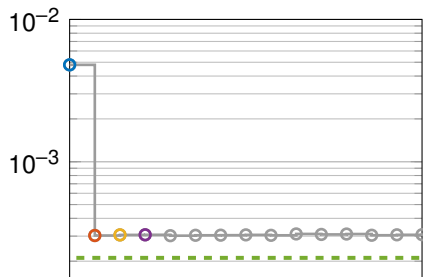


Error 2-norm over iterations:



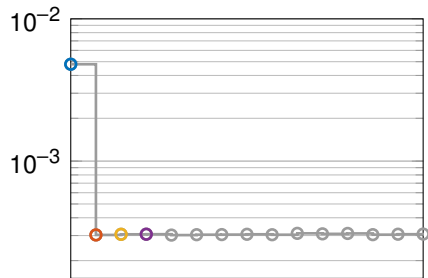


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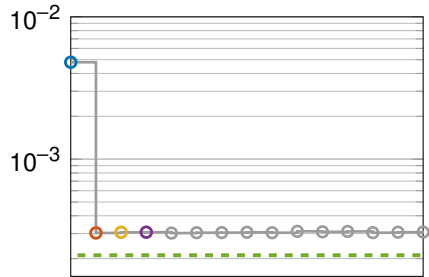


- error norm is a factor 1.5 above the noise level!

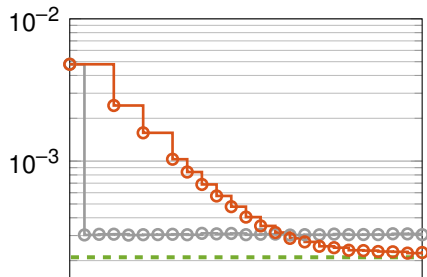
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- ▶ Iteration-varying disturbances: amplified up to a factor 2
- ▶ Trade-off: limiting the amplification by reducing the learning gain α leads to slow convergence (—)



ILC system: $e_j = S(y_d - v_j) - Jf_j, \quad f_{j+1} = Q(f_j + \alpha L e_j)$

Assume

- ▶ y_d is iteration-invariant
- ▶ v_j is iteration-varying
- ▶ $|Q(1 - \alpha L J)| < 1$ (convergence)

Then at iteration j :

$$f_{j+1} = \underbrace{\sum_{i=1}^j (Q(1 - \alpha L J))^{i-1} Q \alpha L S y_d}_{\text{straightforward part}} - \underbrace{\sum_{n=1}^j (Q(1 - \alpha L J))^{j-n} Q \alpha L S v_j}_{\text{all past noise realizations!}}$$

ILC system: $e_j = S(y_d - v_j) - Jf_j, \quad f_{j+1} = Q(f_j + \alpha L e_j)$

Spectrum ϕ_{e_∞} of converged error for $j \rightarrow \infty$:

$$\phi_{e_\infty} = \left| \frac{1 - Q}{1 - (Q(1 - \alpha LJ))} \right|^2 \phi_{S y_d} + \left(1 + \frac{|\alpha J Q L|^2}{1 - |Q(1 - \alpha LJ)|^2} \right) \phi_{S v}$$

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Spectrum ϕ_{e_∞} of converged error for $j \rightarrow \infty$ with $L = J^{-1}$ and $Q = 1$

$$\phi_{e_\infty} = \left(1 + \frac{\alpha^2}{2\alpha - \alpha^2}\right) \phi_{Sv}$$

Key point: we can change learning gain α to influence ϕ_{e_∞}

- ▶ For $\alpha = 1$ the error spectrum $\phi_{e_\infty} = 2\phi_{Sv}$: factor 2 amplification
- ▶ For $\alpha \rightarrow 0$ the error spectrum $\phi_{e_\infty} \rightarrow \phi_{Sv}$: best achievable error

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Spectrum ϕ_{e_∞} of converged error for $j \rightarrow \infty$ with $L = J^{-1}$ and $Q \neq 1$

$$\phi_{e_\infty} = \left| \frac{1 - Q}{1 - (Q(1 - \alpha))} \right|^2 \phi_{S y_d} + \left(1 + \frac{|\alpha Q|^2}{1 - |Q(1 - \alpha)|^2} \right) \phi_{S v}$$

- Reducing $\alpha \in (0, 1]$ increases $\left| \frac{1 - Q}{1 - (Q(1 - \alpha))} \right|^2$ and increases contribution $\phi_{S y_d}$

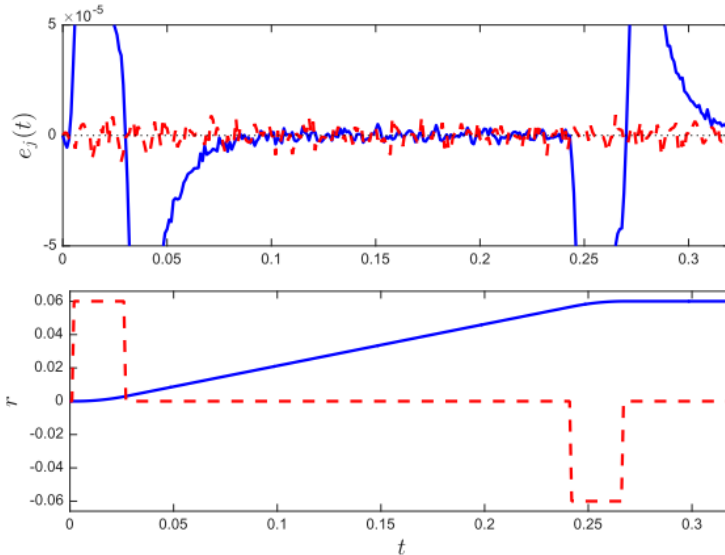
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Choose $\alpha \approx 0$ for varying disturbances, and $\alpha \approx 1$ for repeating disturbances



Error and reference signals wafer stage application

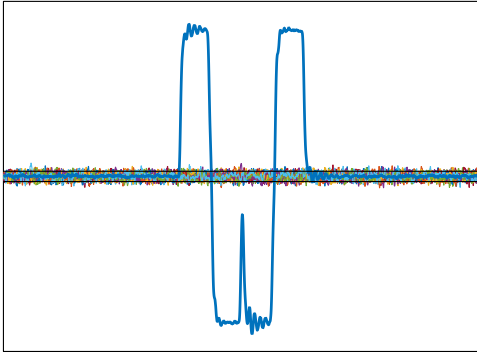
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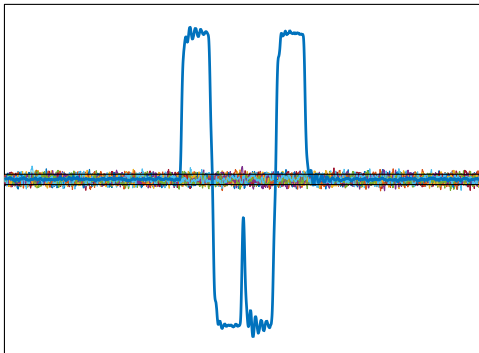
Bottom: reference
acceleration

Learning: $\alpha \approx 0$ between
0.1 and 0.24, $\alpha \approx 1$
elsewhere

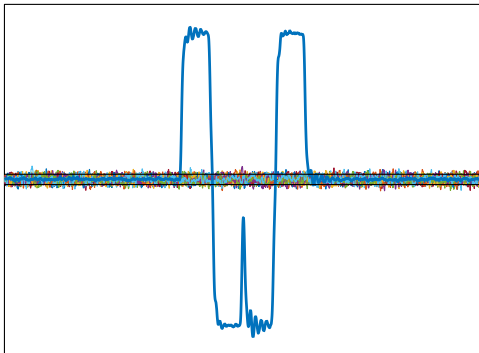
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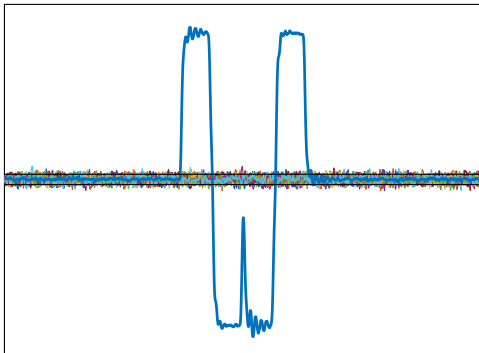
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- ▶ Good performance when $\alpha \approx 0$ for varying disturbances and $\alpha \approx 1$ for repeating disturbances
- ▶ Main idea: filter measured error through a deadzone nonlinearity
- ▶ Result: different learning gains based on signal amplitude

Nonlinear ILC update:

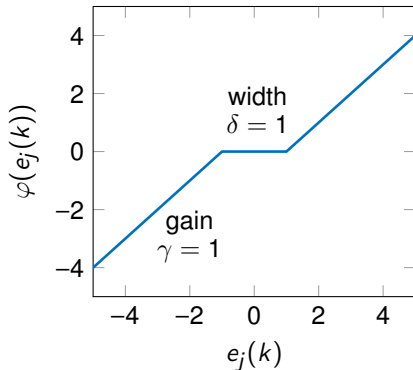
$$f_{j+1} = Q(f_j + \alpha L e_j + L \varphi(e_j))$$

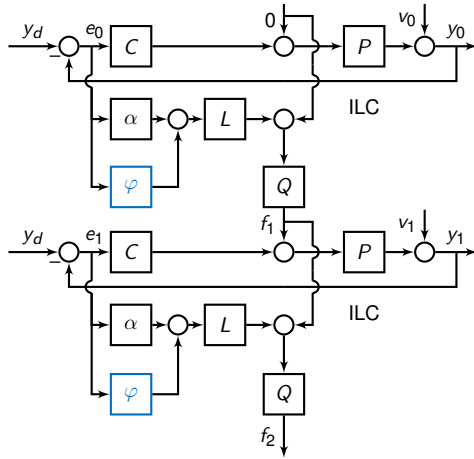
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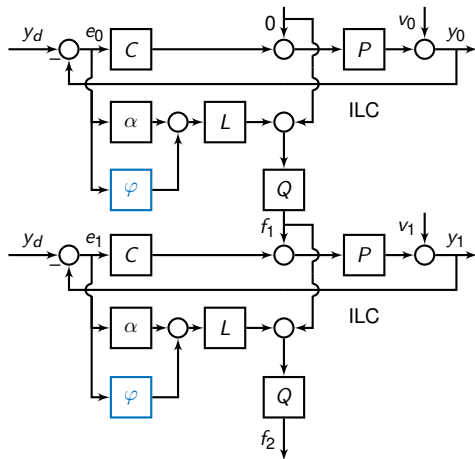
$$f_{j+1} = Q(f_j + \alpha L e_j + L \varphi(e_j))$$

Deadzone nonlinearity $\varphi(e_j)$ is applied to each element k of e_j :

$$\varphi(e_j(k)) = \begin{cases} 0, & \text{if } |e_j(k)| \leq \delta \\ \left(\gamma - \frac{\gamma\delta}{|e_j(k)|}\right) e_j(k), & \text{if } |e_j(k)| > \delta. \end{cases}$$

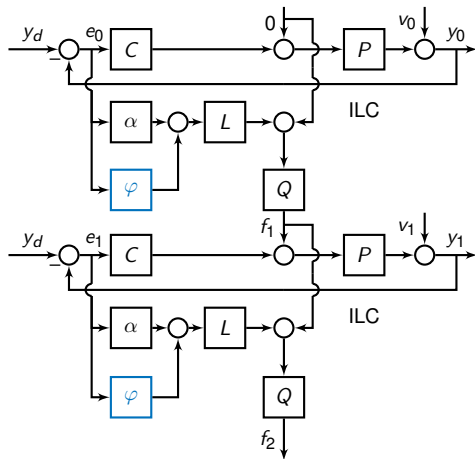






- Resulting update:

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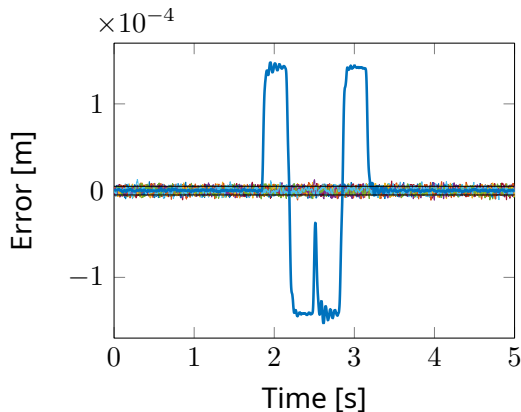


- Resulting update:

$$f_{j+1} = Q(f_j + \alpha L e_j + L \varphi(e_j))$$
- Main idea: choose linear gain α small and nonlinear gain γ large

- $n_e = 20$ experiments with $f_j = 0$

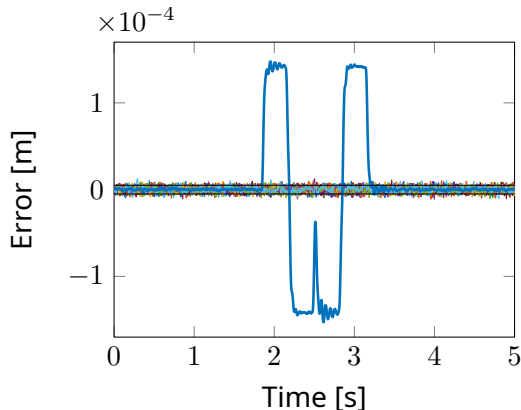
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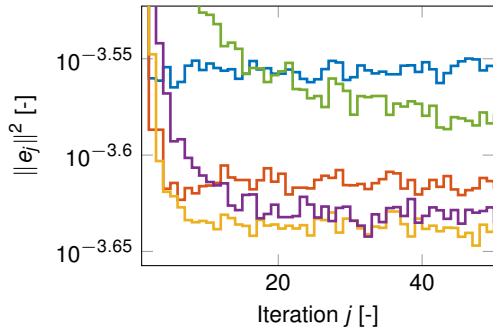
$$\widehat{Sy_d} = \frac{1}{n_e} \sum_{j=0}^{n_e-1} e_j, \quad \widehat{Sv_j} = \widehat{Sy_d} - e_j$$

- Choose δ such that $\varphi(\widehat{Sv_j}) \approx 0$
- Here $\delta = 5 \times 10^{-6}$ (—)



- ▶ The deadzone width δ does not influence convergence/stability

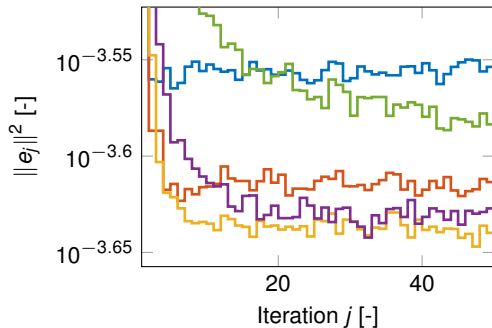
- ▶ The deadzone width δ does not influence convergence/stability
- ▶ However, the performance does depend strongly on δ



Deadzone width: 1×10^{-6} ,
 3×10^{-6} , 5×10^{-6} , 7×10^{-6} ,
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We can find the optimal deadzone width automatically!



Deadzone width: 1×10^{-6} ,
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Main idea: gradient descent step at each iteration:

$$\delta_{j+1} = \delta_j - \beta \frac{\partial \mathcal{J}(\delta)}{\partial \delta}, \quad \text{with} \quad \mathcal{J}(\delta) = \hat{\mathbf{e}}_{j+1}^T(\delta) \hat{\mathbf{e}}_{j+1}(\delta)$$

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- ▶ Error estimate: $\hat{\mathbf{e}}_{j+1}(\delta) = \hat{\mathbf{S}}_{y_d} - \hat{\mathbf{J}} f_{j+1}(\delta)$
- ▶ $\hat{\mathbf{S}} = (1 - \hat{\mathbf{P}}\mathbf{C})^{-1}$ and $\hat{\mathbf{J}}$ are available from ILC design procedure
- ▶ Reference y_d is known

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- ▶ $\hat{\mathbf{S}} = (1 - \hat{\mathbf{P}}\mathbf{C})^{-1}$ and $\hat{\mathbf{J}}$ are available from ILC design procedure
- ▶ Reference y_d is known
- ▶ Gradient expression: $\frac{\partial \mathcal{J}(\delta)}{\partial \delta} = \left(-2\hat{\mathbf{J}}^T \hat{\mathbf{e}}_{j+1} \right)^T \frac{\partial f_{j+1}}{\partial \delta}$

The derivative $\frac{\partial f_{j+1}}{\partial \delta}$ follows from rewriting the deadzone expression:

$$\frac{\partial f_{j+1}}{\partial \delta} = QL\varphi_2(e_j, \delta), \quad \text{with } \varphi_2(e_j(k), \delta) = \begin{cases} 0, & \text{if } |e_j(k)| \leq \delta \\ -\gamma \text{sign}(e_j(k)), & \text{if } |e_j(k)| > \delta. \end{cases}$$

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Then, the gradient of criterion $\mathcal{J}$ with respect to δ is given by

$$\frac{\partial \mathcal{J}(\delta)}{\partial \delta} = \left(-2\hat{\mathbf{J}}^\top \hat{\mathbf{e}}_{j+1}(\delta) \right)^\top QL\varphi_2(\mathbf{e}_j, \delta).$$

At each ILC iteration:

1. Estimate $\hat{e}_{j+1}(\delta) = \hat{S}_{y_d} - \hat{J}f_{j+1}(p_\varphi)$ through simulation with $\hat{S}$ and $\hat{J}$

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5. Compute ILC update $f_{j+1} = Q(f_j + \alpha L\hat{J}e_j + L\varphi(\delta_{j+1}, e_j))$

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2. Design $L \approx J^{-1}$ (stable inversion, ZPETC)
3. Design zero-phase lowpass filter Q such that $\|Q(1 - JL)\|_{\mathcal{L}_\infty} < 1$ (linear case with $\alpha = 1$)
4. Choose $\gamma \approx 1$, $\alpha \approx 0$ and $\alpha + \gamma \approx 1$ while satisfying

$$\left\| Q \left(1 - \alpha JL - \frac{\gamma}{2} JL \right) \right\|_{\mathcal{L}_\infty} + \frac{\gamma}{2} \|QJL\|_{\mathcal{L}_\infty} < 1$$

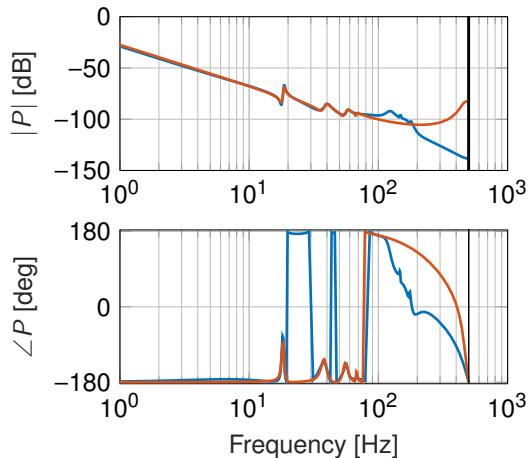
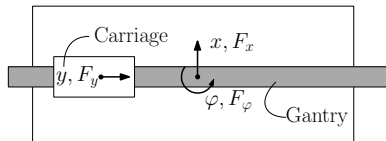
1. Determine deadzone width δ based on experiments without ILC, or use self-optimizing approach:
2. Design $L \approx J^{-1}$ (stable inversion, ZPETC)
3. Design zero-phase lowpass filter Q such that $\|Q(1 - JL)\|_{\mathcal{L}_\infty} < 1$ (linear case with $\alpha = 1$)
4. Choose $\gamma \approx 1$, $\alpha \approx 0$ and $\alpha + \gamma \approx 1$ while satisfying

$$\left\| Q \left(1 - \alpha JL - \frac{\gamma}{2} JL \right) \right\|_{\mathcal{L}_\infty} + \frac{\gamma}{2} \|QJL\|_{\mathcal{L}_\infty} < 1$$

Note similarity of convergence condition to linear case $\implies$ design directly using Bode plots!

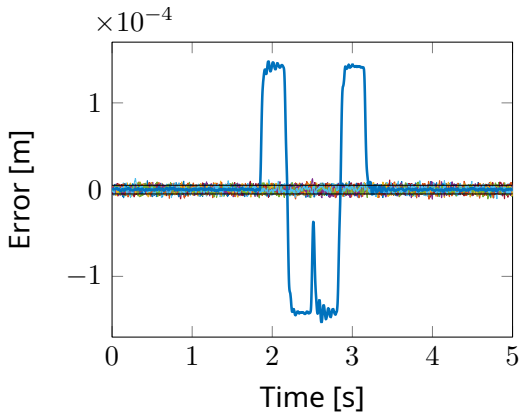
1. Nonlinear ILC: main idea
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4. **Simulation example**
5. If time permits: convergence/stability for nonlinear ILC/RC

- y -translation of the carriage of a flatbed printer
- System (—) and low-order approximation (—)



- $n_e = 20$ experiments with $f_j = 0$

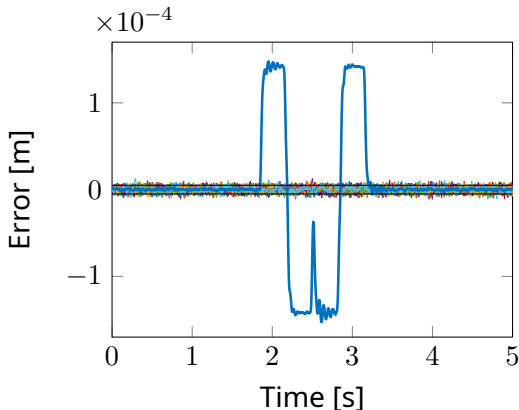
$$\widehat{Sy_d} = \frac{1}{n_e} \sum_{j=0}^{n_e-1} e_j, \quad \widehat{Sv_j} = \widehat{Sy_d} - e_j$$



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$$\widehat{Sy_d} = \frac{1}{n_e} \sum_{j=0}^{n_e-1} e_j, \quad \widehat{Sv_j} = \widehat{Sy_d} - e_j$$

- Choose δ such that $\varphi(\widehat{Sv_j}) \approx 0$
- Here $\delta = 5 \times 10^{-6}$ (—)
- Alternative: self-optimizing approach, only choose initial δ_1

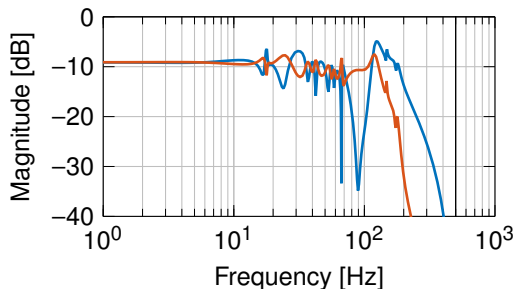


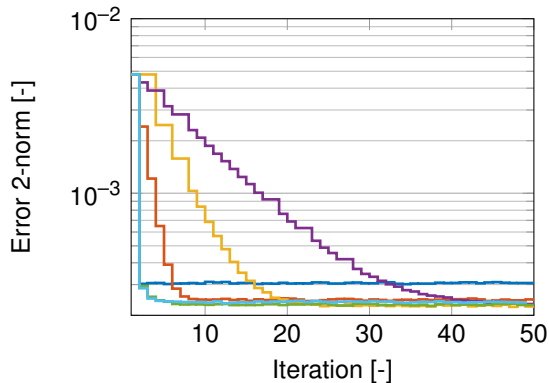
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$$\|Q(1 - JL)\|_{\mathcal{L}_\infty} < 1$$
(linear case with $\alpha = 1$)

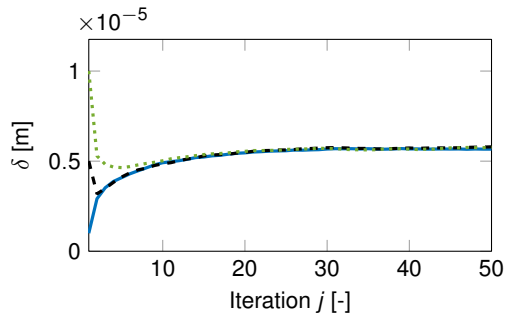
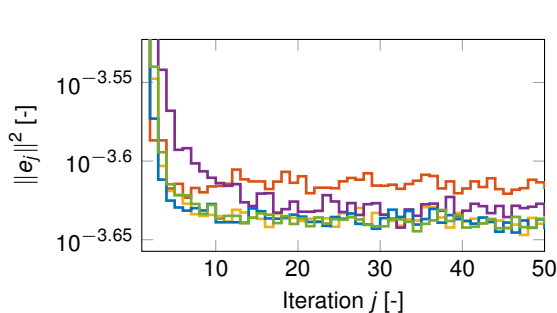
1. Choose δ using 20 experiments without feedforward
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4. Choose $\gamma \approx 1$, $\alpha \approx 0$ and $\alpha + \gamma \approx 1$ while satisfying

$$\underbrace{\left\| Q \left(1 - \alpha JL - \frac{\gamma}{2} JL \right) \right\|_{\mathcal{L}_\infty}}_{=0.572(\text{for } \alpha=0.3, \gamma=0.7)} + \underbrace{\frac{\gamma}{2} \|QJL\|_{\mathcal{L}_\infty}}_{=0.422} < 1$$





- Nonlinear ILC removes the trade-off between fast convergence and limited amplification of varying disturbances



Deadzone widths:

- Constant: 3×10^{-6} , 5×10^{-6} , and 7×10^{-6}
- Self-optimizing: $\delta_1 = 1 \times 10^{-6}$, $\delta_1 = 5 \times 10^{-6}$ and $\delta_1 = 1 \times 10^{-5}$
- For different initial values, the self-optimizing deadzone width performs as well as the constant, optimal value

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- ▶ Solution: use a deadzone in nonlinear ILC to apply different learning gains based on amplitude characteristics
- ▶ Leads to fast convergence and small errors
- ▶ Further reading and related work:
 - ▶ Also works for lifted ILC [Aarnoudse, Pavlov and Oomen (Automatica 2025)] and repetitive control
 - ▶ Method and design procedure in [Aarnoudse, Pavlov and Oomen (CEP 2024, CDC 2025)]

1. Nonlinear ILC: main idea
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Theorem: Monotonic convergence of nonlinear frequency-domain ILC

The sequence of iterates $\{e_j\}$ is monotonically convergent in the 2-norm for $v_j = 0 \forall j$ if

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- ▶ $\mathcal{L}_\infty$ -norms can be evaluated using measured frequency-response data
- ▶ Intuitive tuning of robustness filter Q using Bode diagrams

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$$0 \leq \frac{\varphi(a) - \varphi(b)}{a - b} \leq \gamma$$

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3. Ensure that there exists $\rho < 1$ for which

$$\|\mathcal{F}(\mathbf{e}_1) - \mathcal{F}(\mathbf{e}_2)\|_2 \leq \rho \|\mathbf{e}_1 - \mathbf{e}_2\|_2$$

- ▶ [Aarnoudse, Pavlov and Oomen (CEP 2024)] Leontine Aarnoudse, Alexey Pavlov and Tom Oomen, "A design framework for nonlinear iterative learning control and repetitive control: Applied to three mechatronic case studies," *Control Engineering Practice*, vol. 149, pp. 105976, 2024. DOI:10.1016/j.conengprac.2024.105976
- ▶ [Aarnoudse, Pavlov and Oomen (Automatica 2025)] Leontine Aarnoudse, Alexey Pavlov and Tom Oomen, "Nonlinear iterative learning control for discriminating between disturbances," *Automatica*, vol. 171, pp. 111902, 2025. DOI:10.1016/j.automatica.2024.111902
- ▶ [Aarnoudse, Pavlov and Oomen (CDC 2025)] Leontine Aarnoudse, Alexey Pavlov and Tom Oomen, "Self-optimization of nonlinear iterative learning control and repetitive control," *Proceedings of the 63rd IEEE Conference on Decision and Control*, pp. 765-770, Rio de Janeiro, Brazil, 2025
- ▶ [Oomen and Rojas (Mechatronics 2017)] Tom Oomen and Cristian Rojas, "Sparse iterative learning control with application to a wafer stage," *Mechatronics*, vol. 47, pp. 134-147, 2018. DOI:10.1016/j.mechatronics.2017.09.004